
WHAT ARE MATRICES - REALLY?

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1 Discovery

Matrices are *linear transformations*: in other words, they input a vector and output a new vector. For example, consider the matrix

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

When we input the vector $\vec{v} = x\hat{i} + y\hat{j}$ or

$$\vec{v} = \begin{pmatrix} x \\ y \end{pmatrix}$$

the matrix outputs

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ y \end{pmatrix}$$

Essentially, we are taking every point (x, y) in the plane to $(2x, y)$, stretching the plane horizontally by a factor of 2.

In general, the matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

sends $\vec{v}$ to

$$\begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}$$

This is precisely why it is called a linear transformation.

If we work in $\mathbb{R}^n$ (veterans would know that this is a *vector space*), there is a unique matrix corresponding to every linear transformation.

Exercise 1.1. Find a matrix to make the following transformations.

- (a) Rotate the plane by 90° clockwise.
- (b) Stretch the x -axis by a factor of a and y -axis by a factor of b .
- (c) Reflect in the x -axis.

2 Multiplication

Now, let's discuss the most common sticking point: matrix multiplication. When we multiply the matrices A and B , we are trying to find the unique matrix AB such that

$$A(B\vec{v}) = (AB)\vec{v}$$

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$$

Then,

$$B\vec{v} = \begin{pmatrix} px + qy \\ rx + sy \end{pmatrix}$$

and

$$A(B\vec{v}) = \begin{pmatrix} (ap + br)x + (aq + bs)y \\ (cp + dr)x + (cq + ds)y \end{pmatrix}$$

which is the same as applying the matrix

$$AB = \begin{pmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{pmatrix}$$

Exercise 2.1. Try to find a matrix that reflects about the x -axis and then takes x to $mx + y$.

Exercise 2.2. Do the algebra for a 3×3 matrix (it hurts a little) and hence convince yourself that multiplication of 3×3 matrices makes sense.

This also helps us see why multiplication isn't commutative. For instance, rotating by 90° and then stretching horizontally gives a different result from stretching horizontally first and then rotating by 90° .

3 Further Reading

In this handout, we have discussed the bare bones of *linear algebra*. If you want to explore, check out part IV of Evan Chen's *An Infinitely Large Napkin*, or Sheldon Axler's *Linear Algebra Done Right*.